INCH-POUND

MIL-C-48162B (AR)
AMENDMENT 2
16 October 1996
SUPERSEDING
AMENDMENT 1
04 November 1992

MILITARY SPECIFICATION

CARTRIDGE, IGNITION, M702
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of MIL-C-48162B (AR), dated 14 August 1987, and is approved for use by the U.S. Army Armament, Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, Add new military specification as follows:

"MIL-P-60312 - Parts, Molded, Plastic Foam, Polystyrene (For use with ammunition)"

PAGE 2

2.1.2, Product and Packaging Drawings, 9287826, delete "Fiberboard" and substitute "Wirebound".

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* 4.4.2.2, Notes, change:

From: "Test shall be performed on material ... eight hour production".

To: "*Test shall be performed on material ... each production shift".

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* 4.4.2.5, Add critical 2 as follows:

"2. Propellant Weight (after loading) ... 100% ... 3.2 ... 4.4.3.1.1/4.5.4.1".

* Major 101, Delete in its entirety.

* Major 102, Delete in its entirety.

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4.4.2.6, Critical 3, add note (a) after sample size and in notes section which reads:

"(a) This Five Foot Drop Test on 25 samples will only be performed on each new lot of primers when they are initially introduced into M702 production."

* Add critical 4 as follows:

"4. Firing plug not loose after assembly ...100% ...3.2 ...Manual"

* Add critical 5 as follows:

"5. Propellant weight ...** ...3.2 ...4.4.3.1.2/4.5.4.2"

* Special A, Delete in its entirety.

* Special B, Delete in its entirety.

* Major 103, Delete in its entirety and substitute:

"103. Propellant weight ...** ...3.5 ...4.4.3.1.2/4.5.4.2"

Major 104, change as follows and add note (b):

"104. Leak test ...60(b) ...3.5 ...4.5.3"

"(b) Test shall be performed on material selected from each days production. If more than 3 failures occur, that days production shall be 100% screened.

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* 4.4.2.6, Delete: Defect 109 - Marking missing, misleading or unidentifiable 0.40% ...3.2 ...Visual

Add: Defect 202 - Marking missing, misleading or unidentifiable 0.65% ...3.2 ...Visual

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4.4.2.7, Add new major and note (a) and (b) as follows:

"102 Moisture content excessive (a) (b) 5.1 (a) (b)

"(a) Requirement and inspection procedure shall be per MIL-P-60312. Two (2) samples shall be randomly selected from a quantity of trays for scheduled production for five (5) consecutive shifts. If one or more samples are defective, the quantity of production represented by the sample shall be rejected. Inspection shall be accomplished within 24 hours of scheduled use of the trays."

"(b) Moisture content determination not required when this packing material is stored in an environmentally controlled area maintained at 40 percent maximum relative humidity (RH) @ 72 degrees F for at least 24 hours prior to using."

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4.4.2.8, Add new major and note (a) as follows:

"103 Moisture content excessive for any packaging component (4 items) (a) 5.1 Gage"

"(a) Components shall be inspected prior to use every hour during the production shift. If any sample exceeds the requirement, the inspection lot shall be rejected from further processing."

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4.4.2.9, Add new major and note (a) as follows:

"104 Environmental conditions inadequate (a) 5.1 Gage"

"(a) Temperature and humidity shall be checked at the beginning and end of each shift. If conditions are not proper at either time, the quantity of product represented by the conditioning shall be rejected for further processing."

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4.4.2.11, 4.4.2.12, 4.4.2.13, add new paragraphs as follows:

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Box, Wirebound, Before Packing			9294951
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMEN PARAGRAPH	NEXT HIGHER ASSEMBLY 9287826 / INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Inside dimensions incorrect	0.40%	5.1	Gage
102	Broken, split or missing components	0.40%	5.1	Gage
Minor				
201	Evidence of poor workmanship	0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Packed Wirebound Box, Before Closing			
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMEN PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Improperly packed	0.40%	5.1	Visual
Minor				
201	Evidence of poor workmanship	0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Packed Wirebound Box After Closing			
CATEGORY	EXAMINATION OR TEST	AQL OR 100%	REQUIREMEN PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Wire closing improper	0.40%	5.1	Visual
Minor				
201	Contents loose	0.65%	5.1	Manual
202	Marking incomplete, incorrect or illegible	0.65%	5.1	Visual
203	Evidence of poor workmanship	0.65%	3.10	Visual

NOTES:

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* 4.4.3.1.1, Delete in its entirety and substitute:

"4.4.3.1.1 Prior to assembly of cap. (See dwg. 9280553). The propellant charge shall be volumetrically dispensed (or tare-weighed) in a container, transferred to the ignition cartridge and then check-weighed 100 percent. The check-weighing shall be as specified in paragraph 4.5.4.1. The volumetric dispensing (or initial weighing) and check-weighing shall be performed by different operators on different scales. Weight limits for weighing shall be established within the applicable requirements, depending upon the accuracy and precision of the weighing system to assure compliance with the charge weight requirements. Any charge of bulk propellant failing to meet the applicable requirements shall be classed defective and removed from the lot."

* 4.4.3.1.2, Delete in its entirety and substitute:

"4.4.3.1.2 After assembly of cap. (See dwg. 9280553). Critical defect - more than 8 grains above assessed maximum weight: Critical defect - more than 8 grains below assessed minimum weight: Major defect - 1 to 8 grains above assessed maximum weight: Major defect - 1 to 8 grains below assessed minimum weight. 315 sample ignition cartridge assemblies shall be randomly selected from each lot for this test. If any sample exceeds the assessed maximum weight by more than 8 grains or if two or more samples exceed the assessed maximum weight by 1 to 8 grains, the lot shall be rejected. If any sample is under the assessed minimum weight by more than 8 grains or if two or more samples are under the assessed minimum weight by 1 to 8 grains, the lot shall be rejected. Testing shall be as specified in paragraph 4.5.4.2."

* 4.4.3.1.3, Delete in its entirety.

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* 4.4.3.2.3, Delete first sentence in its entirety and substitute:

"Twenty-five assemblies consisting of randomly chosen primed bodies, heads and firing plugs shall be selected for this test."

4.4.3.2.3, Add to last sentence as follows:

"(Test shall be performed on new primer lots prior to use in M702 production. Acceptable results shall qualify the primer lot for use in M702 production)."

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* 4.5.1, Delete the test in its entirety and substitute:

"The moisture content of the black powder pellets shall be determined in accordance with the procedures specified in MIL-P-223. A sample of 25 black powder pellets shall be randomly selected from each production shift. If any pellet fails to comply with the requirement cited in paragraph 3.3 herein, all pellets in the production shift represented by this sample shall be rejected. If loading with propellant has begun, ignition cartridges loaded with the non-conforming black powder pellets shall be rejected."

* 4.5.2.3, Delete in its entirety and substitute:

"4.5.2.3 Safety drop test. Each of the test assemblies will consist of an inert filled M720 body, and inert M935 Fuze and a M27 Tailfin assembly. After inserting the sample M702 to be tested, the cartridge total drop weight will be 3.75 pounds +/- 0.05 pounds. The cartridge will be dropped five feet onto a steel plate first in the fin down orientation and then dropped fuze end down. Each cartridge will have the inert M935 fuze and M27 Tailfin assembly replaced after each double drop. The drop height shall be measured from the lowest point of the cartridge orientation to the top of the anvil surface. Any test assembly which functions as a result of the drop shall be classed defective. All questionable test assemblies shall be downloaded to verify primer functioning or nonfunctioning. All components used for this test shall not be returned to their respective lots."

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4.5.3, Delete first sentence in its entirety and substitute:

"After the ignition cartridges have been assembled and the RTV is applied, the ignition cartridges will be kept in the controlled environmental storage for two (2) hours minimum prior to performing the leak test."

* 4.5.4, Delete in its entirety and substitute:

"4.5.4 Weight of propellant."

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* Add paragraph 4.5.4.1 as follows:

"4.5.4.1 Prior to assembly of cap. The ignition cartridge subassembly shall be weighed before and after the propellant is loaded. The weight of the propellant charge shall be determined by subtracting the tare weight of the ignition cartridge subassembly from the gross weight (after loading). The contractor may elect to perform the 100 percent check-weighing operation with automated equipment and mandatory process controls. In such cases, process controls plans in accordance with data item description DI-P1604 must be submitted to the technical agency for approval of the automated operations."

* Add paragraph 4.5.4.2 as follows:

"4.5.4.2 After assembly of cap. The propellant shall be removed from each ignition cartridge sample, placed on the pan of a precision or analytic balance and weighed. Alternatively, the ignition cartridge assembly may be weighed and reweighed after removal of the propellant: The weight of the propellant charge shall be determined by subtracting the weight of the empty ignition cartridge from the loaded (gross) weight. The weight of the propellant in each sample shall be determined to the nearest 0.01 grain."

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